INCH-POUND
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DETAIL SPECIFICATION

CHAIN, STUD LINK, ANCHOR, FLASH BUTT WELDED, NONMAGNETIC

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single shots (90-foot lengths) of ¾-, ⅞-, 1-, 1⅛-, and 1¼-inch, nonmagnetic, flash butt welded, stud link, anchor chain for use on mine warfare ships. Only one classification of chain is covered by this specification.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

COMMERCIAL ITEM DESCRIPTIONS

A-A-59316 - Abrasive Materials; for Blasting

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-22262 - Abrasive Blasting Media Ship Hull Blast Cleaning

MIL-PRF-23236 - Coating Systems for Ship Structures

MIL-PRF-24635 - Coating Systems, Weather-Resistant, Exterior Use

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4010

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

MIL-STD-2142 - Magnetic Silencing Characteristics, Measurement of (Metric)

DOD-STD-2143 - Magnetic Silencing Requirements for the Construction of Nonmagnetic Ships and Craft (Metric)

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING, INC. (ASNT)

SNT-TC-1A - Recommended Practice No. SNT-TC-1A: Personnel Qualification and Certification in Nondestructive Testing

(Copies of this document are available online at www.asnt.org.)

ASTM INTERNATIONAL

ASTM A342/A342M - Standard Test Methods for Permeability of Weakly Magnetic Materials

ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM A388/A388M - Standard Practice for Ultrasonic Examination of Steel Forgings

ASTM E381 - Standard Method of Macroetch Testing Steel Bars, Billets, Blooms, and Forgings

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10012 - Measurement Management Systems – Requirements for Measurement Processes and Measuring Equipment

(Copies of this document are available online at www.iso.org.)

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NACE INTERNATIONAL

SSPC-SP 1 - Solvent Cleaning

SSPC-SP 10/NACE No. 2 - Near-White Metal Blast Cleaning

(Copies of these documents are available online at www.nace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), samples of chain shall be subjected to first article inspection (see 6.5) in accordance with 4.2.

3.2 Material. Form 1 material (unaged, bar, rod, and forging) in the hot finished condition and in accordance with QQ-N-286 shall be used in manufacturing the chain links and studs. The material used in normal production shall be the same composition as the material used in the first article sample in chemistry, quality, and properties. The chain material from each mill run shall be identified through the chain fabrication process. Unidentified material shall be considered nonconforming. Material properties shall be as specified in table I. Column A refers to bar stock material used in making the chain after receiving a heat treatment identical to the chain heat treatment. Column B refers to material specimens taken from a typical chain link.

TABLE I. Mechanical properties after heat treatment.

Property	**Column A bar stock (minimum values)**	**Column B link materials (minimum values)**
Ultimate tensile strength (ksi)	130	120
Yield strength (ksi)	75	70
Elongation (percent)	20	15
Reduction in area (percent)	32	24
NOTE: 1. Kilopound per square inch (ksi).		

3.3 Link dimensions. Link dimensions and properties shall be as specified in table II and on figure 1 (see 6.2). Tolerances shall be as specified in table III and on figure 2. Any flash trim overage existing after the flash butt weld is deburred shall be as specified in table III. The grip radius (G) of each end of the link shall be symmetrical to ensure proper fit and function in the chain handling equipment.

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TABLE II. Stud link chain dimensions and properties.

Chain size (inches)	Length A (inches)	Length over 6 links L (inches)			Link width B (inches)	Number of links per shot	Proof test load (pounds)	Break test load (pounds)	Minimum weight per 15-fathom shot (pounds)	Maximum weight per 15-fathom shot (pounds)
		Minimum	Nominal	Maximum						
¾	4½	19⅜	19½	19⅝	2 11/16	359	39,600	64,800	520	563
⅞	5¼	22⅝	22¾	23 1/16	3⅛	305	53,900	88,200	705	765
1	6	25⅞	26	26⅜	3⅝	267	70,400	115,200	920	1,000
1⅛	6¾	29 1/16	29¼	29⅝	4 1/16	237	89,100	145,800	1,165	1,270
1¼	7½	32 5/16	32½	32 15/16	4½	213	110,000	180,000	1,435	1,565

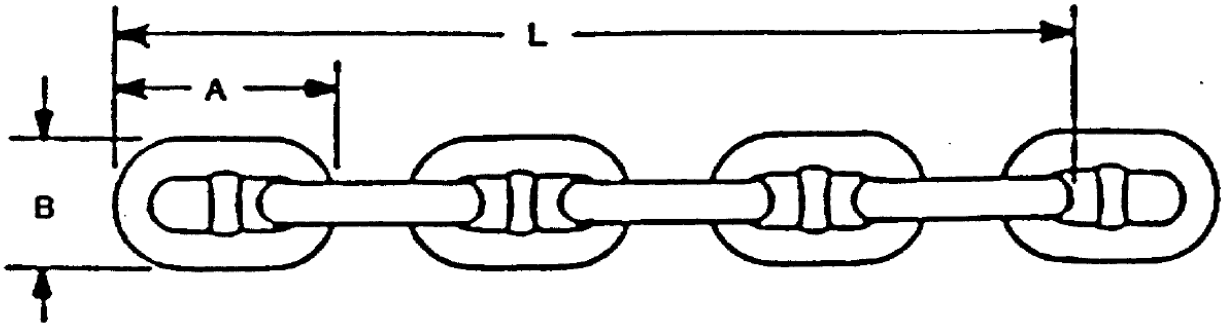


FIGURE 1. Stud link chain dimensions.

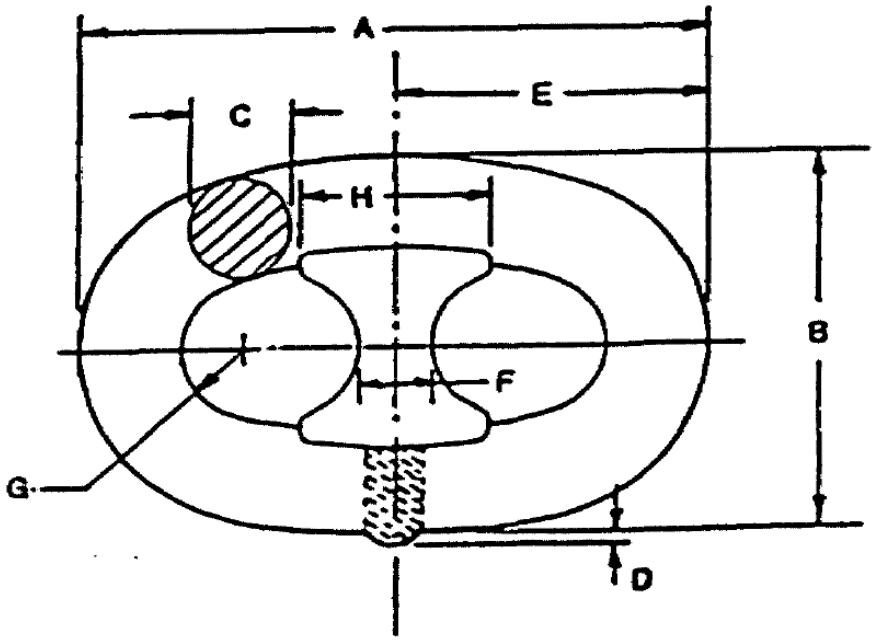
TABLE III. Stud link tolerances.

Nominal chain size (C) (inches)	Permissible tolerances (inch)							
	Link length (A) plus or minus	Link length (B) plus or minus	Wire diameter ovality (maximum) 1/	Maximum flash trim overage (D)	Stud centering plus or minus	Stud width		Grip radius (G) plus or minus
						(F) plus	(H) minus	
¾ to 1¼ (inclusive)	1/16	1/16	1/32	1/16	3/16	1/16	⅛	1/16

NOTE:

1/ When the link diameter (dimension C) is checked for ovality (major axis minus minor axis), the area of the elliptical cross section (π/4 × major axis × minor axis) shall be at least equal to the area of a circle of the chain size (π /4 × link diameter²). Dimension C shall be measured at a point generally as specified on figure 2.

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E = 0.5 × A
F = 0.75 × C
G = 0.85 × C
H = 2.0 × C

FIGURE 2. Basic link and stud dimensions.

3.4 Link and stud surface. Burrs, irregularities, and rough edges shall be contour ground to a fair surface. Links and studs shall also be free of cracks, surface irregularities, dents, and undercutting in excess of 1/16 inch when examined as specified in 4.4.2 and 4.6.3. Attention shall be given to inspection in the following areas:

a. Grip area while links interface during loading.
b. Region gripped by clamping dies during flash welding.
c. Flash weld buildup perimeter after trimming.

Any ridge, flashing, or other projection in these areas shall be ground flush or blended with the fair link surface.

3.5 Flash butt weld.

3.5.1 Weld surface. The link surfaces at the flash weld shall be free of cracks, lack of fusion, and gross porosity when examined with liquid penetrant as specified in 4.4.3.

3.5.2 Fusion of weld joint. The flash weld shall be free of defects causing ultrasonic (UT) back reflections equal to or greater than those produced from a 0.05-inch surface notch calibration standard when tested as specified in 4.4.4.

3.5.3 Fusion zone microstructure. The heat-affected zones on each side of the weld joint shall be free of cracks when examined as specified in 4.4.8.

3.6 Stud size and alignment. Stud size and shape, alignment, and centering shall be as specified on figure 2 (see 4.4.7 and 4.6.12) and shall have no gaps visible between the stud and the link body at the ends of the H dimension. Non-continuous gaps are permissible along the length H.

3.6.1 Stud penetration. Prior to being pressed into the link body, every stud shall be free from unwanted protrusions and flash resulting from manufacturing or forging operations. Stud penetration (at midplane) into the link body resulting from the stud pressing process shall be so that a firm stud fit results; that is, there shall be no detectable looseness in any direction and there shall be no more than 0.05C (C is the nominal chain size) maximum depth on either side of the link. Stud penetration shall be measured as specified in 4.4.5. The perimeter of the stud saddle shall be radiused (sharpness removed) to prevent cutting into the link body.

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3.7 Link and stud interior. Links and studs shall be free from harmful defects such as laps, seams, pipes, cracks, fins, porosity, nonmetallic inclusions, and segregation when examined as specified in 4.4.5 and 4.6.13.

3.8 Chain length. The chain shall be in 90-foot shots (15 fathoms) and consist of all common links, the quantity of which is specified in table II (see 6.2). The overall length of six consecutive links measured from every third link shall be within the limits specified in table II (see 4.4.1). By this method of measurement, every link, except the three links at each end of the chain, shall be measured twice with different combinations of adjacent links totaling six links (see 4.6.1).

3.9 Chain weight. The weight of each shot of chain shall be in accordance with table II. The required weight limits of six-link samples shall be calculated from the information in table II. Weighing shall be done as specified in 4.4.10.

3.10 Chain breaking load. The chain shall withstand the breaking load specified in table II and shall not crack in the flash butt weld, or the flash butt weld heat-affected zone, when tested as specified in 4.5.2.

3.11 Chain proof load. The chain shall withstand the proof load specified in table II when tested as specified in 4.5.3.

The actual load applied shall:

a. Not cause cracking, fracture, or deformation.

b. Not permanently increase chain length more than 1.7 percent or cause the chain to exceed the six-link and individual link dimensions.

c. Be used on a full shot or any portion thereof.

d. Not be applied more than three times.

e. Be indicated on strip charts along with the resulting elongation.

3.11.1 Chain sizing. Chain shall be sized (lengthened) to meet six-link length requirements by applying loads greater than the proof load (see 4.5.3 and 4.6.1).

3.11.2 Maximum sizing load. Sizing loads greater than 120 percent of the proof test load shall be cause for rejection of the entire shot (where a full shot is sized in one operation) or the affected links (where fractions of a shot are sized in continuous production processes).

3.12 Chain properties. Sample links detached from finished chain shall be sectioned and tested for determination of properties and bending characteristics (see 4.5.4, 4.5.5, 4.5.7, and 4.4.9). Mechanical properties shall be as specified in table I. During bend testing, cracking or fracture shall not occur until the bend angle reaches 120 degrees when tested as specified in 4.5.5. The magnetic permeability (mu) of finished links shall be in accordance with DOD-STD-2143 and shall not exceed 1.2 mu in accordance with MIL-STD-2142, test method P01. Each shot shall be tested in accordance with MIL-STD-2142.

3.13 Standards of manufacture. Manufacture of the chain shall be as specified in 3.13.1 through 3.13.5. Equipment and processing parameters used to manufacture the first article inspection samples shall also be used for production chain.

3.13.1 Heating and bending. Bars shall be heated before bending using electric resistance heating. Peak bar temperatures shall not exceed 2,100 degrees Fahrenheit (°F). Bending should be performed with bar temperatures above 1,600 or below 900 °F. Reheating of the bar is permitted, but prolonged soaking at 2,100 °F shall be avoided.

In order to prevent undue scaling or flaking of the bar due to excessive heat, the heating phase shall be automatically controlled by an optical heat sensor for electric resistance heating. The controller shall be checked for calibration at least once every 8 production hours.

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3.13.2 Flash butt welding. The bent bar shall be electrically flash butt welded. Maximum bar end misalignment shall not exceed 5 percent of the bar diameter. The bar ends shall be parallel within 5 degrees. Bar end misalignment shall be checked a minimum of once every 10 links.

The following welding parameters shall be controlled during welding of each link:

a. Platen motion.

b. Current as a function of time.

c. Hydraulic pressure.

d. Flashing voltage.

After removal of excess flash butt weld material, a visual inspection of each link in the area to be covered by the stud shall be performed in accordance with 4.4.

3.13.3 Heat treatment. Heat treatment shall be performed after welding has been completed and prior to testing and inspection. No more than two complete heat treatments (annealing plus age hardening) shall be permitted. A system of identification shall be established to ensure that no shot or portion thereof will be heat treated more than twice. Temperatures and times, or temperatures and chain speed, shall be controlled. After exit from the furnace, the links shall be cooled in such a manner that differential cooling does not occur relative to opposite sides of the links.

3.13.3.1 Annealing. Annealing of the chain after welding may be used to refine grain size and equalize internal stresses after fabrication and welding on the links. It shall be performed at 1,800 to 1,875 °F (in order to produce a uniform temperature within the links) and followed by a water quench to room temperature.

3.13.3.2 Age hardening. Aging of the chain shall be performed to obtain a combination of strength and other mechanical properties such as ductility.

3.13.4 Markings. The stud of each link shall be permanently marked in raised letters or letters indented with a blunt nose tool. One side shall be marked for the U.S. Navy (for example, “USN”), along with chain size and nonmagnetic material identification (for example, “1¼ NM”). The other side shall identify the manufacturer or trade name of the chain. The studs of both end links of 90-foot shot shall be marked with serial numbers assigned by the manufacturer and traceable to production, test inspection, and certification records. Markings shall be a minimum of 5⁄32 inch in height for ¾-inch to 1⅛-inch chain, and ¼ inch in height for 1¼-inch chain. Markings shall be raised or indented a minimum of 1⁄32 inch. An area for serial number marking may be prepared by grinding.

3.13.5 Finishing. Surfaces shall be cleaned in accordance with SSPC-SP 1 to remove oil, grease, and residual contaminants. Even traces of residual oil will degrade paint adhesion. Safety precautions for working with flammable solvents shall be enforced. Surfaces shall be abrasive blasted to a level of surface cleanliness in accordance with SSPC-SP 10/NACE No. 2 using an abrasive in accordance with MIL-A-22262 or A-A-59316. Surface profile shall be 2 to 4 mils. After completion of all tests and inspections, the chain shall be painted with either two coats of paint in accordance with MIL-PRF-24635, type V or VI, class I, at a dry film thickness (DFT) of 5 to 8 mils per coat; or one coat of paint in accordance with MIL-PRF-23236, type VI at 4 to 8 mils DFT topcoated with one coat of paint in accordance with MIL-PRF-24635, type V or VI, class I, at 4 to 8 mils DFT.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on samples of chain when a first article sample is required (see 3.1). This inspection shall consist of the examinations and tests specified in table IV.

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TABLE IV. First article inspection.

Inspection	Requirement	Examinations and tests	Sample	Rejection and retest
Chain proof load	3.11	4.5.3	4.2.1.a	4.6.7
Chain length	3.8 and 3.11.1	4.4.1	4.2.1.a	4.6.1 and 4.6.2
Chain weight	3.9	4.4.10.1	4.2.1.a	4.6.5
Link dimensions	3.3	4.4.6	4.2.1.c	4.6.12
Stud size and alignment	3.6	4.4.7	4.2.1.c	4.6.12
Stud penetration	3.6.1	4.4.5	4.2.1.b	4.6.13
Chain breaking load	3.10	4.5.2	4.2.1.c	4.6.6
Link and stud surface	3.4	4.4.2	4.2.1.a	4.6.3
Flash weld surface	3.5.1	4.4.3	4.2.1.a	4.6.4
Flash weld fusion	3.5.2	4.4.4	4.2.1.a	4.6.10
Flash weld microstructure	3.5.3	4.4.8	4.2.1.b	4.6.13
Link interior (macrostructure)	3.7	4.4.5	4.2.1.b	4.6.13
Chemical composition	3.2	4.5.1	4.2.1.b	4.6.14
Chain specimen tensile test	3.12	4.5.4	4.2.1.b	4.6.8
Chain specimen bend test	3.12	4.5.5	4.2.1.b	4.6.9
Chain specimen hardness [1/]	3.12	4.5.7	4.2.1.b	-----
Chain magnetic permeability	3.12	4.4.9	4.2.1.c	4.6.11
Bar stock tensile test	3.2	4.5.6	4.2.1.d	4.6.8
Flash butt welding	3.13.2	4.4	4.2.1.a	4.6.3.1

NOTE:
[1/] For information only.

4.2.1 First article inspection samples. The following samples shall be prepared for the first article inspection. All samples shall be thoroughly cleaned of loose scale and welding slag and proof tested prior to performing the following examinations and tests:

a. One six-link section for nondestructive testing (see table IV).

b. Single links for destructive testing (quantity as required in table IV).

c. One triplet, minimum, (from [a] above) for breaking load test (see 4.5.2), for examination of link and stud dimensions (see 4.4.6), for stud alignment and size (see 4.4.7), and magnetic permeability (see 4.4.9).

d. One (8- to 12-inch) length of bar stock processed through annealing and age hardening as though it were a length of chain. The material shall have been selected from bar stock of the same heat or melt of material as that used for making first article inspection samples.

4.3 Conformance inspection. Bar stock and chain samples shall be selected in accordance with 4.3.1 and shall be inspected as specified in table V. In addition, a detailed inspection of the chain shall be performed following cleaning of all surfaces to remove scale and after proof testing. The chain shall be arranged to permit careful inspection from all sides. The chain shall be bright and free of paint or other coating which would tend to conceal defects during the testing and inspection.

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TABLE V. Conformance inspection (for production chain).

Inspection	Requirement	Examinations and tests	Sample	Rejection and retest
Chain proof load	3.11	4.5.3	4.3.1.3	4.6.7
Chain length	3.8 and 3.11.1	4.4.1	4.3.1.3	4.6.1 and 4.6.2
Link and stud surface	3.4	4.4.2	4.3.1.3	4.6.3
Flash weld surface	3.5.1	4.4.3	4.3.1.3	4.6.4
Flash weld fusion	3.5.2	4.4.4	4.3.1.5	4.6.10
Stud penetration	3.6.1	4.4.5	4.3.1.5	4.6.13
Chain weight	3.9	4.4.10	4.3.1.3	4.6.5
Link dimensions	3.3	4.4.6	4.3.1.4	4.6.12
Stud size and alignment	3.6	4.4.7	4.3.1.4	4.6.12
Chain breaking load	3.10	4.5.2	4.3.1.4	4.6.6
Chain specimen tensile test	3.12	4.5.4	4.3.1.5	4.6.8
Flash weld microstructure	3.5.3	4.4.8	4.3.1.5	4.6.13
Chain magnetic permeability	3.12	4.4.9	4.3.1.4	4.6.11
Bar stock tensile test	3.2	4.5.6	4.3.1.1	4.6.8

4.3.1 Sampling for conformance inspection.

4.3.1.1 Bar stock material samples. For sampling, bar stock material to be used in construction of chain, one 8- to 12-inch long section, removed from either end of the bar stock, shall be selected from each heat or melt of material.

4.3.1.2 Finished chain. For sampling finished chain (cleaned, proof tested, and inspected), four or more extra links shall be provided with each shot for destructive and nondestructive testing. The extra links and shots shall be processed together and then cut apart to provide proof-tested shots for delivery and detached triplets and links for testing.

4.3.1.3 Shots. Chain proof load testing (see 4.5.3) and examination of chain length (see 4.4.1 and 4.4.6) shall be conducted on each shot of chain. The surface examination of links (see 4.4.2) and flash butt welds (see 4.4.3) shall be conducted after proof load testing. The weight of finished shots shall be determined (see 4.4.10).

4.3.1.4 Detached triplets. A minimum of one triplet from each shot shall be removed for the chain breaking load test (see 4.5.2). Dimensions of each link of the triplet shall be measured for conformance to link dimensions (see 4.4.6); grip radius, stud size, and alignment (see 4.4.7); and magnetic permeability (see 4.4.9) prior to the breaking load test.

4.3.1.5 Links. Extra links for each shot shall be removed in a manner that allows specimens to be taken for the chain specimen tensile test (see 4.5.4), flash butt weld fusion zone microstructure (see 4.4.8), interior examination of flash butt weld fusion (see 4.4.4), and stud penetration (see 4.4.5).

4.3.1.6 Disposition of chain samples. Unless otherwise specified (see 6.2), all chain test samples and unused samples shall be scrapped. If retained or shipped, each sample shall be identified according to the shot from which it was taken.

4.4 Examinations.

4.4.1 Chain length. Chain length measurements shall be made after proof testing. The chain shall be free of twist and under a tension of 10 percent of proof load. The overall length of six links shall be measured from every third link of each shot for conformance to 3.8.

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4.4.2 Surface examination of link and stud. After cleaning all surfaces to remove scale and after proof testing, all surfaces of each link and stud shall be visually examined for conformance to 3.4.

4.4.3 Surface examination of flash butt weld. Liquid penetrant inspection shall be employed to examine the flash butt weld area of links. Procedures for liquid penetrant inspection shall be in accordance with T9074-AS-GIB-010/271. Acceptance criteria shall be in accordance with MIL-STD-2035. This examination shall be done after cleaning all surfaces to remove scale and after proof testing but before any painting. Every link shall be examined. Flash butt welds shall meet the requirements specified in 3.5 (see 4.6.3 and 4.6.4).

4.4.4 Examination of flash weld fusion. UT examination shall be employed to examine the flash butt weld area of links in accordance with ASTM A388/A388M, except that the frequency of examination shall be every link, by at least a level II nondestructive testing examiner, in accordance with SNT-TC-1A. Links shall be examined for conformance to 3.5 (see 4.6.10). For first article examination, all links of the sample shall be examined.

4.4.5 Examination of link and stud interior (macrostructure). A single link shall be cut in half on a plane parallel with both the long and short axes, etched, and examined (see 4.6.13). The sample shall be one piece or in three pieces from the same link, approximately equal in size, showing (1) bend section, (2) weld and stud penetration section, and (3) the other stud penetration section. The significance of indications observed in the microstructure shall be rated in accordance with ASTM E381. Penetration of the stud into the link body as a consequence of the stud pressing operation shall be determined and shall conform to 3.6.1 and 3.7.

4.4.6 Link dimensions. The dimensions of each link, including grip radii, of each detached triplet (see 4.3.1.4) shall be measured for conformance to 3.3. Dimensions in table II shall be checked when dimensional discrepancies are observed in visual examination of each shot (see 4.4), and as required for links of triplets detached from the finished chain (see 4.6.12).

4.4.7 Stud size and alignment. The size, alignment, and centering of each stud shall be measured on each link of each detached triplet (see 4.6.12) to determine if the requirements of 3.6 are satisfied.

4.4.8 Examination of flash butt weld fusion zone microstructure. A sample of material shall be taken from the link to show the microstructure on a plane cutting the fusion zone perpendicular to the weld joint. The specimen shall be about 1.0 square inch in area and shall be polished and etched for microstructural inspection at 100× magnification for conformance to 3.5.3.

4.4.9 Examination of link magnetic permeability. Triplets removed from the chain breaking load test shall be examined for magnetic permeability prior to other testing. Each sample shall be visually examined with particular attention given to the weld area and stud, using a permeability indicator in accordance with ASTM A342/A342M, and with an appropriate insert calibrated in accordance with ISO 10012 (see 3.12).

4.4.10 Chain weight. The finished chain shall be weighed on completion of marking (see 3.13.4) and finishing operations (see 3.13.5), and after removal of any extra links needed for destructive testing.

4.4.10.1 First article sample weight. For the first article inspection, the sample six-link section shall be weighed (see 4.2.1.a). The minimum and maximum weight of the six-link section shall be calculated from the information listed in table IV. The minimum and maximum shot weights shall be divided by the number of links per shot and the results multiplied by six to establish the minimum and maximum weights for the six-link section.

4.4.11 Marking. The stud of each link shall be examined for conformance to 3.13.4.

4.5 Testing methods.

4.5.1 Chemical composition. The material sample shall be selected in accordance with 4.2.1.b and analyzed for chemical composition (see 3.2) by using chemical or spectrographic analysis techniques. The analysis shall be done independently from the material supplier for verification purposes.

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4.5.2 Chain breaking load test. While completely free from twist, first article and production samples of finished chain shall be subjected to the appropriate tensile load of table II for 15 seconds (minimum) and examined for cracking after the load is removed. The loading arrangement on the end links shall duplicate the arrangement between any other two links. After inspection, the samples for first article testing shall be loaded to destruction to determine ultimate chain strength (for information only). The test triplets shall be disposed of as specified in 4.3.1.6. The break testing machine shall have been calibrated within the 12 months preceding the chain breaking load test (see 4.6.6).

4.5.3 Chain proof load test. While completely free from twist, each chain sample shall be loaded in tension to the value specified in table II. The loading arrangement on the end links shall duplicate the arrangement between any other two links. The proof load testing machine shall have been calibrated in accordance with ISO 10012 within the 12 months preceding the chain proof load test. Applied loads shall not exceed 120 percent of proof test load value nor produce a permanent length increase greater than 1.7 percent.

4.5.4 Chain specimen tensile test. Two tensile specimens shall be taken longitudinally from a test link, one from each side and midway between the ends of the link as shown on figure 3 (see 4.2.1.b and 4.3.1.5). Tensile tests shall be conducted with procedures and using standard specimens machined in accordance with ASTM A370 (see 3.12 and 4.6.8).

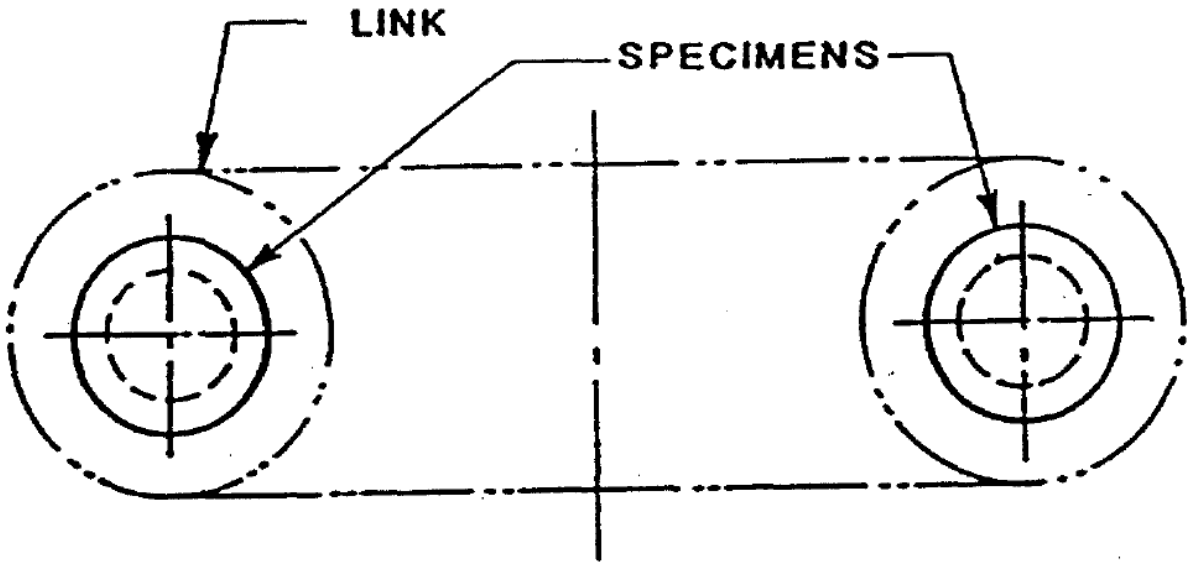


FIGURE 3. Orientation of tensile test specimen.

4.5.5 Chain specimen bend test. Orientation of the flash butt weld bend test specimen shall be as shown on figure 4. The sample shall be taken from an area of the link below the stud (see 4.2.1.b). The test piece shall be prepared with the maximum area of the original bar surface retained and with a maximum thickness of 0.45 inch. The test shall be conducted with the bar surface in tension using a mandrel having a diameter of 4.4×T. The specimen shall be precisely positioned over the mandrel such that the weld area is parallel to and in contact with the mandrel.

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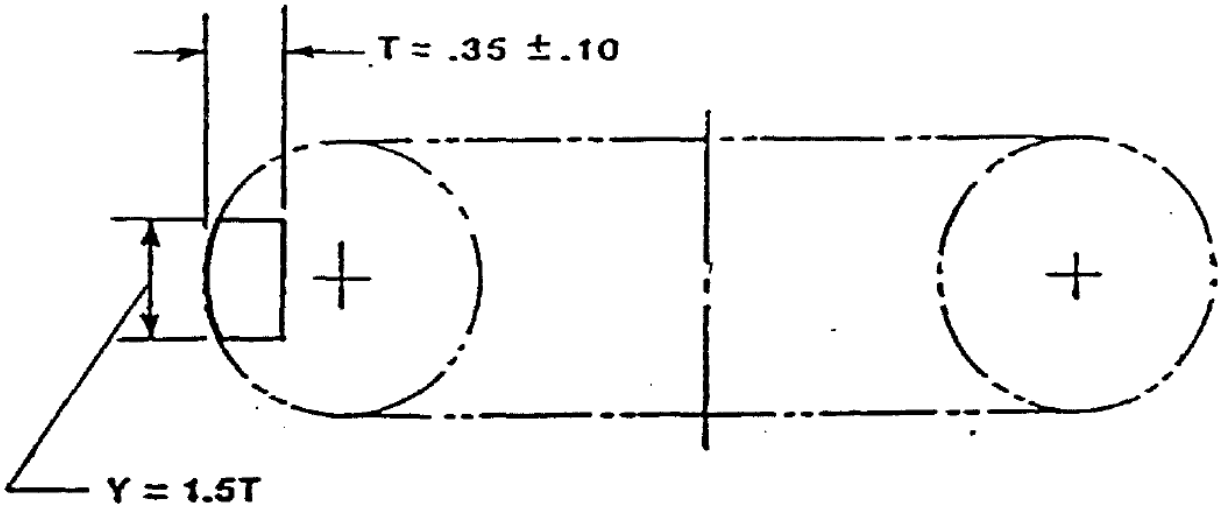


FIGURE 4. Orientation of bend test specimen.

4.5.6 Bar stock tensile test. The specimen shall be taken from the bar stock sample (see 4.2.1.d and 4.3.1.1) and tested as specified (see 4.5.4) in accordance with ASTM A370 (see 4.6.8). The specimen shall then be subjected to the annealing and age hardening treatments to be used for production chain.

4.5.7 Brinell hardness test. The Brinell or Rockwell C scale hardness of a test link (see 4.2.1.b) shall be established on the outside of the link on the side opposite the flash butt weld in accordance with ASTM A370.

4.6 Retest and rejection criteria.

4.6.1 Length. If the length of six consecutive links is short after proof testing, the chain may be stretched by applying loads up to 120 percent of the proof test load specified in table II, provided that the requirements of 3.11 are met. If the chain length over six links exceeds the tolerances shown in table II, the overlength chain links shall be cut out and 4.6.2 shall apply (see 4.4.1), except the first article sample chain, which shall conform to the dimensions shown in table II.

4.6.2 Single links. If single links are found to be defective or not meeting other applicable requirements, defective links shall be cut out and connecting links inserted in their place. After the entire shot has successfully completed all manufacturing and inspection, the connecting link shall be removed and the two resulting lengths may be used to start new shots of chain. No shot or portion thereof shall be heat treated more than twice.

4.6.3 Cracks. If a crack is found visually or by the liquid penetrant test of 4.4.3, it shall be eliminated by grinding to a maximum depth of 1/16 inch and streamlining the surface to eliminate any abrupt discontinuities. The repaired area shall then be examined using the dye penetrant inspection technique in accordance with T9074-AS-GIB-010/271. The area shall be free of cracks. If the defect is still present, 4.6.2 shall apply.

4.6.3.1 Surface irregularities. If a surface irregularity, dent, or undercut of a depth exceeding the criteria of 3.4 is found, this shall be cause for rejection of the link and 4.6.2 shall apply.

4.6.4 Flash butt weld. Linear indications found in the flash butt weld by dye penetrant inspection that are greater than 1/8 inch shall be cause for link rejection and 4.6.2 shall apply. Indications of 1/8 inch or less shall be treated as cracks (see 4.6.3).

4.6.5 Weight. Chain that is not within the weight limits as specified in 3.9 shall be rejected.

4.6.6 Break. If a chain triplet breaks anywhere or cracks in a welded area (see 3.10) below the break test load value specified in table II, the associated shot shall be rejected.

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4.6.7 Proof test failure. If the chain fails to satisfy the requirements of 3.11 during the proof test, 4.6.2 shall apply (see 4.5.3).

4.6.8 Tensile test failure. If the results from tensile testing chain and bar stock specimens fail to meet the requirements in table I but are within 2,000 pounds per square inch (lb/in^2) of the required tensile strength or within 2 percent of the required elongation, a retest of another specimen selected from the same sample will be permitted. If the second test fails to meet the requirements of table I, the shot shall be rejected (see 4.5.4 and 4.5.6).

4.6.9 Bend test failure. If there is cracking or fracture associated with the flash butt weld or in the heat-affected zone of the bend specimen before 120 degrees of bend is made, the first article samples shall be rejected (see 4.5.5).

4.6.10 UT inspection failure. If indications of interior flash butt weld defects exceed the standards of 3.5.2 during UT examination of the flash weld zone, 4.6.2 shall apply (see 4.4.4).

4.6.11 Magnetic permeability. Chain exceeding the permeability limits of 3.12 shall be rejected.

4.6.12 Dimensions. If link diameter, length, width, grip radii, stud, stud alignment, and centering dimensions do not conform to the required dimensions, these shall be compared to the dimensions of other links in the triplet and links of all sample chain representing the shot. If a single particular dimension fails to meet the required dimensional tolerance in more than one-half of the sample links, all links in each shot shall be examined for link dimensions, grip radii, stud, stud alignment, and size. If a single particular dimension fails to meet the required dimensions in five of the links in a shot, the shot shall be rejected and 4.6.2 shall apply (see 3.3, 3.6, 4.4.6, and 4.4.7).

4.6.13 Macrostructure and microstructure. Rejection in macrostructural and microstructural examinations shall be for defects, cracks, or discrepancies in regard to microstructure requirements (see 3.5.3, 3.7, 4.4.5, and 4.4.8). If stud penetration exceeds the limits specified in 3.6.1, the two detached links in the shot shall be checked. If one other link shows unacceptable stud penetration, the shot shall be rejected.

4.6.14 Material composition. Material not within the composition tolerances shall be rejected (see 3.2 and 4.5.1).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The chain covered by this specification is intended to be used for anchoring mine warfare ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.1).
c. Chain size (link diameter) (see 3.3).
d. Number of shots of chain required (see 3.8).
e. Disposition of first article and other samples (see 4.3.1.6).
f. Packaging requirements (see 5.1).
g. Required documentation of the inspection accomplished by surveyor (see 6.6).

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6.3 Certification. To be considered a certified producer of anchor chain and to be eligible to bid on solicitations to this specification, a contractor should currently be approved by the American Bureau of Shipping (ABS) to produce extra-high strength, grade 3a, flash butt welded, stud link anchor chain, and authorized to use the American Petroleum Institute (API) monogram on chain manufactured in accordance with ANSI/API Specification 2F. A contractor bidding on solicitations to this specification should identify the place of manufacture of the chain, the source of rolled bars and the source of forged studs to be used in fulfilling any resulting contract.

6.4 Retention of records. Agencies using this specification to acquisition anchor chain for use on U.S. Navy ships should forward a copy of all data to Naval Sea Systems Command, Deck and Underway Replenishment Systems Machinery Division (SEA 05Z43), Department of the Navy, Washington, DC 20376-0001, for retention.

6.5 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of the number of units specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Classification society inspection. Unless otherwise specified (see 6.2), the contractor should provide the services of a surveyor from a member society of the International Association of Classification societies during the manufacture and testing of chain made to this specification. The credentials and experience of the particular surveyor (by name) should be submitted to and approved by the Government before the first article construction begins. The contractor should maintain an inspection and record-keeping system satisfactory to the surveying society. The surveyor should witness the measurement and testing of all chain.

6.7 Standards of manufacture. Any exceptions for paragraph 3.13 that are considered acceptable or needed to obtain acceptable chain should be identified and submitted for information.

6.8 Existing stocks. Current stock of nonmagnetic anchor chains may be used until depleted; future requisitions for nonmagnetic anchor chains should refer to the current version of this specification.

6.9 Subject term (key word) listing.

Brinell hardness

Chain proof load

Magnetic permeability

Studs

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

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DLA – GS

Preparing activity:
Navy – SH
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